

Meeting Notes

Meeting notes moved to dedicated date tabs.

Mar 4, 2026

Mar 3, 2026

MARCH 3, 2025

- Overall Analysis
 - Jarsmeka
 - Previous Campaign
 - Return customers are likely to resubscribe
 - Customers that said no before is less likely to say yes later
 - Recently contact customers are likely to subscribe
 - A reasonable number of follow-ups help, but too many are harmful
 - Campaigns
 - Best time to run campaigns: September to April
 - Longer phone calls (5+ minutes) increase success
 - A reasonable number (??) of follow-ups help, but too many are harmful
 - Either cell phone or telephone works
 - Demographics
 - Aim to contact individuals with jobs in:
 - blue collar
 - Management
 - Technicians
 - Admin
 - Services
 - Retirees
 - students
 - A more educated user is likely to sign up
 - Divorcees are less likely to sign up
 - Financial Profile
 - Richer clients with account balances over \$1,428 are likely to subscribe - what defines a rich client
 - Clients with existing loans (personal/housing) are less likely to subscribe
 - Credit default clients are less likely to subscribe
 - Housing loans are more common than person loans
 - Aim for high-balance customers with no loans
 - **IDEAL CAMPAIGN:** The campaign should run from September to April. Focus on existing customers, especially those who previously subscribed, as prior subscription is the strongest predictor of future conversion. Target customers who were contacted recently. Use 2–3 follow-ups; more than that reduces effectiveness. Calls should last 5+ minutes, as longer duration increases success. Contact method (cell vs telephone) does not significantly impact results.
 - **IDEAL CLIENT PROFILE:** The ideal customer is financially stable, with an account balance above \$1,428, no personal or housing loans, and no credit default. High-balance customers without debt have the highest conversion rates. Target individuals in blue-collar, management, technician, administrative, service roles,

retirees, and students. Higher education increases the likelihood of subscription.
Divorced customers have lower conversion rates.

- How many clients can be contacted next time based on the above profile?

Action Items for March 3

- ☒ ~~Vince created two new buckets in the raw data~~
- ☐ Kateryna to create a 00_data_cleaning in the notebooks
- ☐ Kateryna will be moving all cleaned EDA to notebooks.
- ☒ ~~Jarsmeka will be moving all cleaned EDA to notebooks.~~
- ☒ ~~Kerensa will be moving all cleaned EDA to notebooks.~~
 - ☐ Format includes - Question, Code, Visualization, Analysis
 - ☐ Add blur to README to help understand organization

Action Items for March 4, due before March 5 meeting

- ☐ **Kateryna** to create a 00_data_cleaning in the notebooks, and upload cleaned data into folder
- ☐ **Kateryna** will be moving all cleaned EDA to notebooks.
- ☐ **Kateryna** and Lanlan to work on model for Campaign Strategy
- ☐ **Kateryna** to create streamlit page with each of our questions: Campaign Strategy
- ☐ **Jarsmeka** to create streamlit page with each of our questions: Financial Profile/Previous Campaign
- ☐ **Kerensa** to create streamlit page with each of our questions: Demographics
- ☐ **Lanlan** and **Kerensa** to create streamlit page for Prediction Model
- ☐ **@all** to review/update 🟡 DSI Showcase - March 2026
 - ☐ **Jarsmeka** (first half) and **Lanlan** (second half) to present on Saturday

For visualizations:

- 1) remember to make graph titles/yaxis/xaxis to be business executive/business team friendly.
- 2) And if there is any color needed for your visuals - lets use these:

- #1F4E52
- #2F6F73 (This one as primary)
- #5E8F95
- #8FB3C1
- #C6D9E2

Feb 26, 2026

Feb 26, 2026 | 📅 DSI: Technical Learning Modules

Attendees: jarsmek@google.com J M

Notes

- Team went over updates
- 📄 Project Plan - DSI Project Team 2
- All work should be moved to scratch folder to name:
https://github.com/DzeDze/ds_team2_bank_marketing/tree/main/scratch
- Role assigned: Vince to work on Segmentation EDA instead Campaign Strategy, Jarsmek to work on showcase slides

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- Vinh-phuc Nguyen: Repo Administration, Campaign Strategy EDA, Visualisation - Streamlit  
- Jarsmek Maneckarase: Financial Profile EDA, Previous Campaign Effectiveness EDA  
- Kateryna Makieieva: Data cleaning, preprocessing, handling missing values, Campaign  
  Strategy EDA, Classification Model,  
- Yuen Yan Wong: Readme file, Project Proposal, Demographics EDA  
- Chris Odetola  
- Lanlan Li
```

Action items

- ☒ ~~Team members will complete all analysis and visualization~~
- ☐ Team will review all pull requests/analysis before meeting again on Tuesday. The goal is to ensure we understand all findings before our next call
- ☐ Kerensa to complete changes to project proposal and submit
- ☐ Vince to create "Document" folder for README and project proposal

Feb 25, 2026

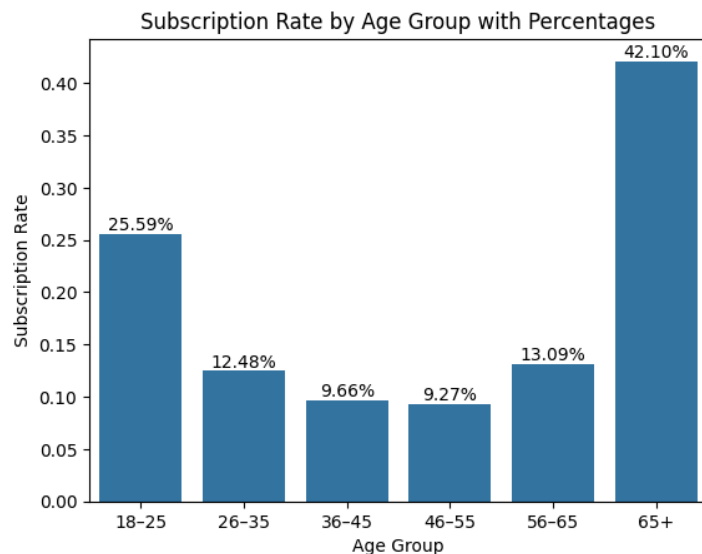
Feb 25, 2026 | Team Project

Notes -

- Business Problem
- Business Motivation
- Intended Audience
- Expected Impact
- Compiled Questions:

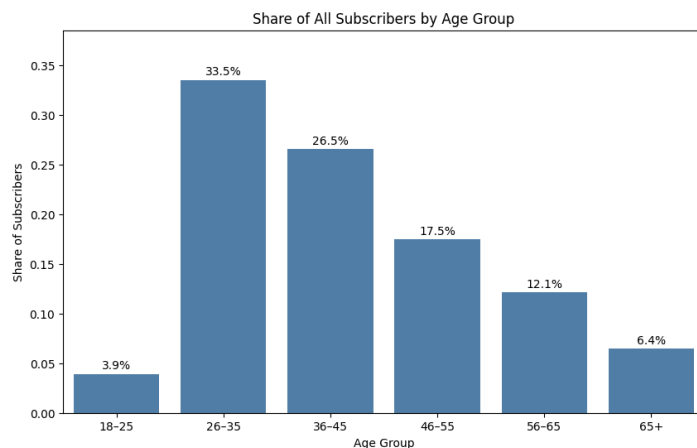
a. 01) Demographics kerensa.ca@gmail.com

i. Which age groups are most likely to subscribe to a term deposit? (Jarsmeka)



(Likelihood within each age group ← finding which

groups more likely to subscribe)



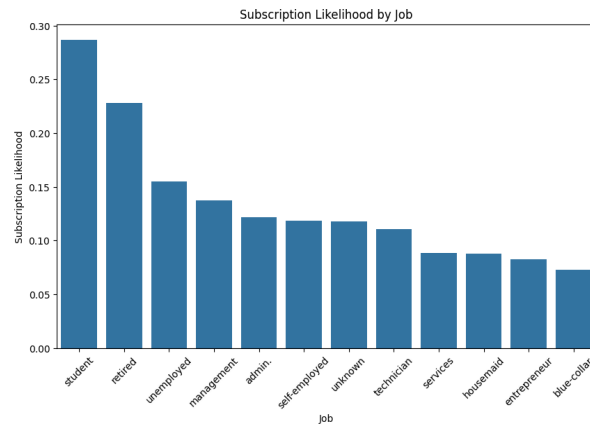
(Volume - 26-45age = most subscriber coz they

are the largest group, but their subscription rate is low (not responsive) ← above graph)

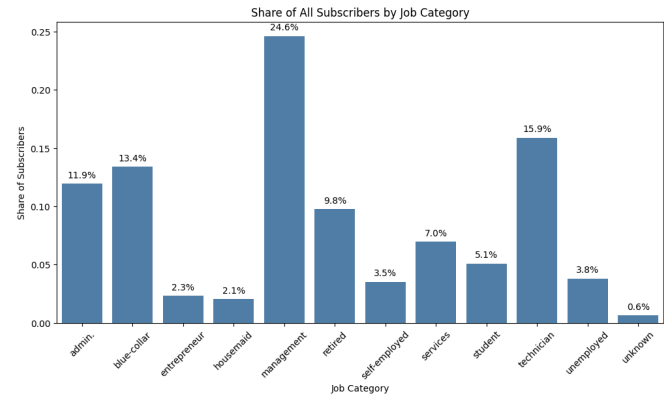
**** 65+ → high conversion rate/potential; 18-25 → responsive; 26-45: large group but low responsiveness)**

ii. Are retired customers more likely to subscribe than working professionals? (Vince)

Likelihood:



Volume:

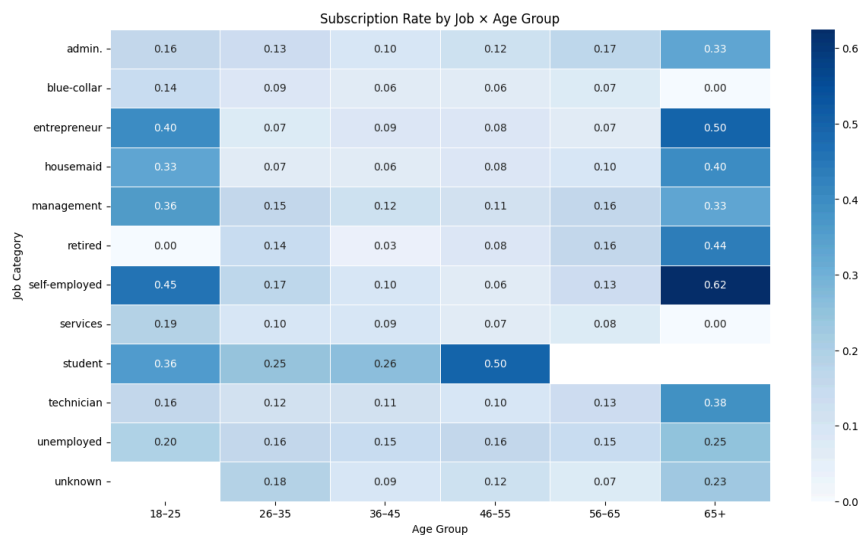


- Student and retired are easiest to convert, but they are small groups
 - Great to target, high ROI campaign
- Volume mostly come from working professionals (management, technician, blue-collar, admin)
 - Low likelihood, but large group so even small improvement in conversion can produce big gains
- High likelihood $\neq$ high volume (same as age group result)

iii. Does a customer's job influence subscription probability? (Vince, Jarsmeka)

Age vs job analysis

Heatmap (subscription likelihood for every job-age combination)

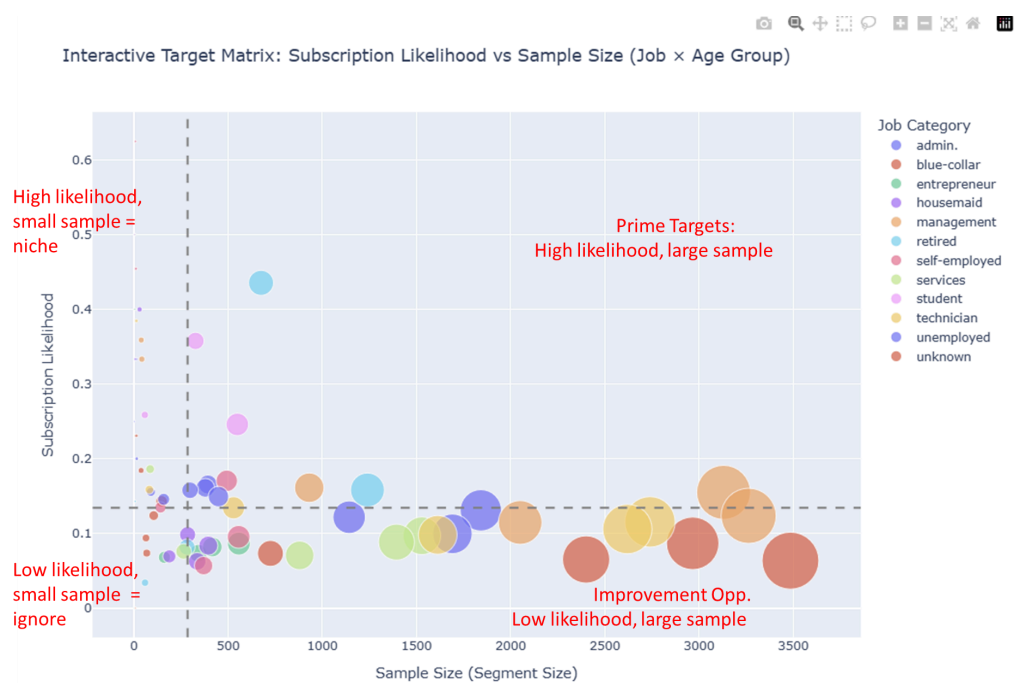
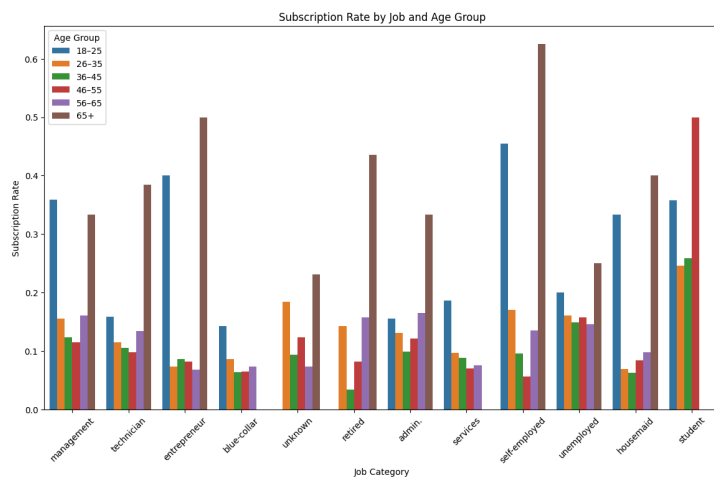


- self-employed(65+) highest subscription rate among all, retired (65+) also high
- student (18-45): high likelihood in both young and mid-age group
- blue-collar: low across all age group $\leftarrow$ job effect dominate age effect
- management: young age and 56+ has moderate likelihood
- Entrepreneur & Housemaid: relatively high likelihood (18-25 & 65+)

** some job (incl. Student, retired, self-employed) has high propensity
Blue-collar (in all age) & service: low propensity **

sample_size = df.pivot_table(						
age_group	18-25	26-35	36-45	46-55	56-65	65+
job						
admin.	90	1842	1691	1143	393	12
blue-collar	147	2967	3486	2401	725	6
entrepreneur	5	341	557	416	162	6
housemaid	6	188	336	395	285	30
management	39	3131	3264	2051	931	42
retired	1	7	59	282	1240	675
self-employed	11	493	556	370	141	8
services	86	1529	1394	880	264	1
student	327	549	58	4	0	0
technician	82	2740	2620	1613	529	13
unemployed	15	379	449	298	158	4
unknown	0	38	64	105	68	13

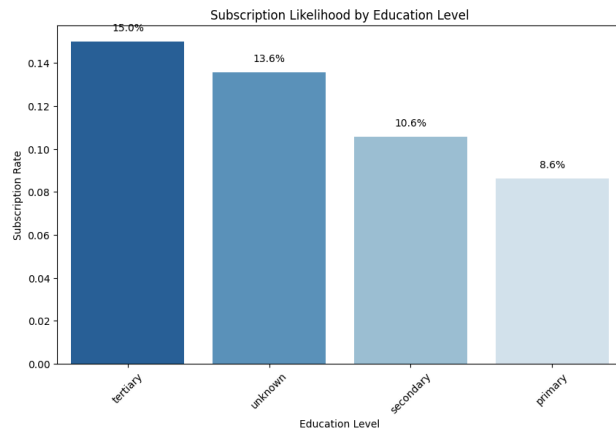
Grouped Barplot (Volume)



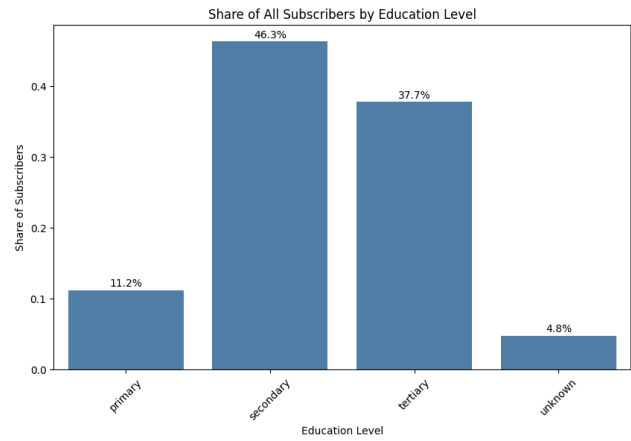
(use median to separate)

iv. Does a customer's education level impact subscription? (Jarsmeka)

Likelihood:



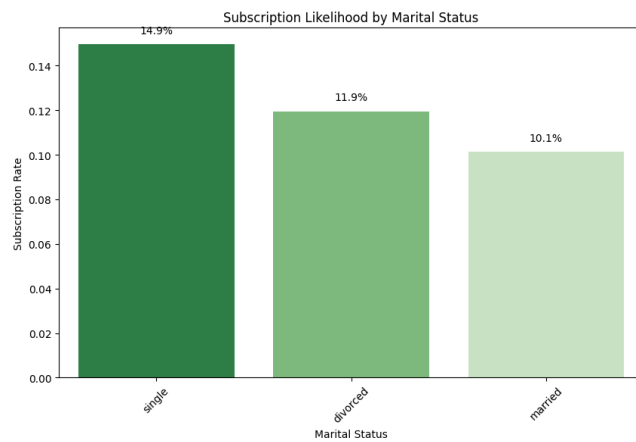
Volume:



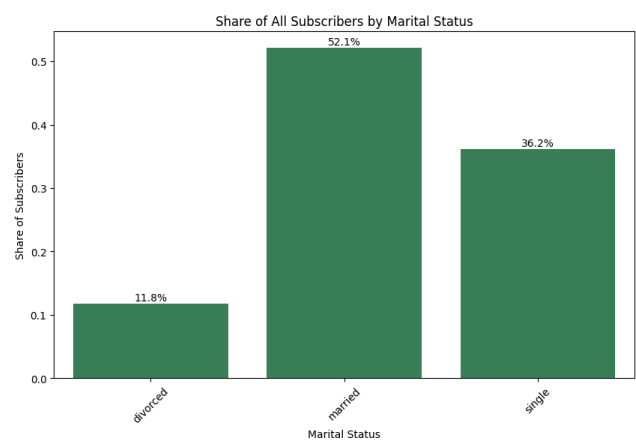
- Tertiary: highest likelihood
- Secondary: highest volume

v. How does marital status impact subscription behavior? (Jarsmeka)

Likelihood:

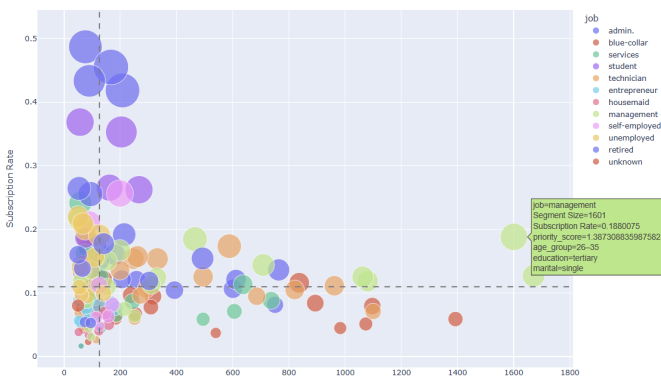


Volume:



- Single: highest likelihood
- Married: highest volume

High-Value Segments Across Age × Job × Education × Marital



(all 4 factors)

- a. **02) Financial Profile** **J M - COMPLETED - keep all**
 - i. Does higher account balance increase likelihood of subscription? (Vince, Jarsmekka)
 - ii. Does having a housing loan affect subscription probability? (Vince)
 - iii. Does having a personal loan affect subscription behavior? (Vince)
 - iv. Does having credit in default reduce subscription probability? (Vince)
 - v. Is there an interaction between balance and loan status? (Vince)
 - vi. Can subscription likelihood be linked to other products (e.g., loans)? (Kerensa)
 - vii. How many loans are personal vs housing loans? (Jarsmekka)
- b. **03) Campaign Strategy** **@Kateyna**
 - i. Does contact type (cellular vs telephone) affect subscription success? (Jarsmekka)
 - ii. Is there an optimal number of contacts in a campaign? (Vince)
 - iii. Does subscription probability decrease after too many calls? (Vince)
 - iv. What months have the highest vs lowest subscription success? (Jarsmekka, Vince)
 - v. Are certain days of the week more effective? (Vince)
 - vi. Is there evidence of seasonal patterns in subscription behavior? (Vince)
 - vii. How does call duration relate to subscription? (Vince)
- c. **04) Previous Campaign Effectiveness** **J M - COMPLETED - all**
 - i. How does the outcome of a previous campaign affect current success? (Vince, Jarsmekka)
 - ii. Are customers with previous successful campaigns more likely to subscribe again? (Vince)
 - iii. Are customers previously unsuccessful unlikely to convert again? (Vince)
 - iv. Does time since last contact (pdays) influence success probability? (Vince)
 - v. Are existing customers more likely to subscribe than new ones? (Kerensa)
- d. **05) Segmentation & Predictions** **phuc.nguyen.ufo@gmail.com**
 - i. What combination of age, job, balance, and loan status yields the highest conversion rate? (Vince)
 - ii. Can we identify low-probability segments to avoid targeting? (Vince)
 - iii. Can customers be ranked by likelihood of subscription? (Vince)
 - iv. Does combining variables provide stronger predictive power than single variables? (Kerensa)
 - v. Does the combination of age + job + loan status improve prediction compared to single factors? (Kerensa)
- Risks and Unknowns
 - a. **Risks** **kerensa.ca@gmail.com**
 - i. The dataset may be heavily skewed toward “no” subscriptions compared to “yes” (class imbalance). (Vince, Jarsmekka)
 - ii. The dataset only includes campaigns from a Portuguese bank, which may limit applicability in other countries (e.g., Canada). (Kerensa, Jarsmekka)
 - iii. The data is older than 2013, so customer behavior and economic conditions may have changed (e.g., financial crises, COVID), reducing predictive relevance today. (Jarsmekka, Kerensa)
 - iv. Class imbalance may affect model accuracy and interpretation of results. (Vince)

- v. Data quality concerns: “unknown” values in variables such as job, education, contact, and outcome. (Kerensa)
- vi. The “other” category in outcome lacks explanation and may reduce interpretability. (Kerensa)
- vii. Extreme outliers (e.g., very high balances or very long call durations) may skew results. (Kerensa)
- viii. Predictive models may perform well on historical data but fail in real-world future campaigns. (Kerensa)

b. Unknowns

- i. The cost of running each marketing campaign. (Jarsmeka)
- ii. The profit generated per successful subscription. (Jarsmeka)
- iii. Why customers choose not to subscribe (behavioral or psychological reasoning). (Jarsmeka)
How long customers maintain their term deposit after subscribing. (Jarsmeka)
- iv. Whether loans are actively being used or simply held. (Jarsmeka)
- v. Other non-demographic factors influencing subscription (e.g., personal motivations). (Jarsmeka)
- vi. Whether the campaign strategy or call scripts changed over time. (Jarsmeka)
- vii. External economic factors that may have influenced customer decisions. (Vince)
- viii. Important variables not included in the dataset (e.g., income, digital engagement, geographic location). (Kerensa)
- ix. Time of day of calls (morning vs evening). (Kerensa)
- x. Whether customers consented to repeated contact (ethical consideration). (Kerensa)
- xi. Correlation does not imply causation — observed relationships may not represent true drivers. (Kerensa)

Action items

- ☐ Analysis and Visualization to be completed by Friday
 - ☐ Initial Analysis, Demographics and Project Proposal - Kerensa
 - ☐ Financial Profile and Previous Campaign Effectiveness, Initial Showcase Slideshow - Jarsmeka
 - ☐ Campaign Effectiveness and Segmentation/Prediction - Vince